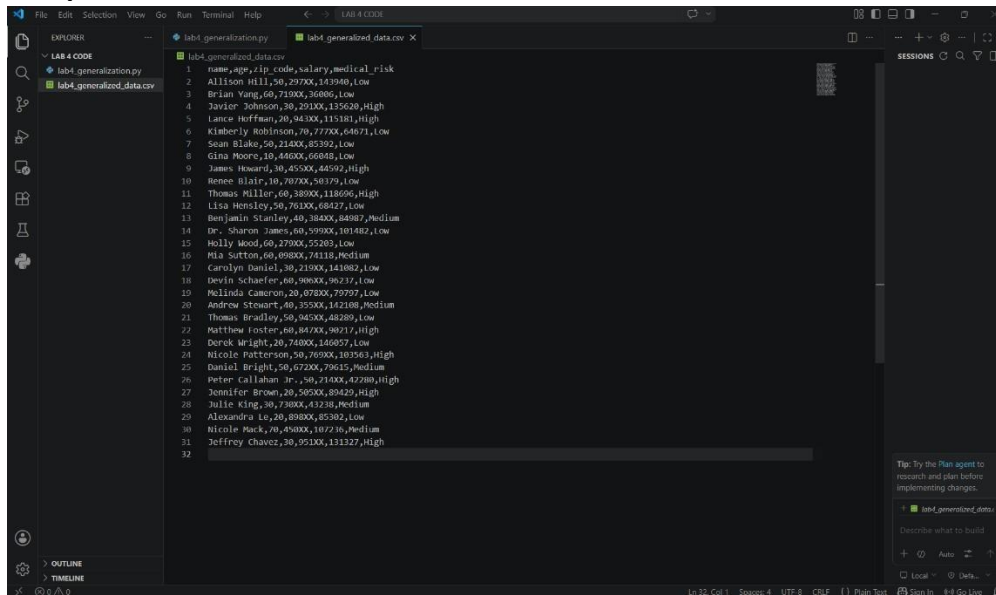


INFOE 3 LAB ACTIVITY 4 GENERALIZATION

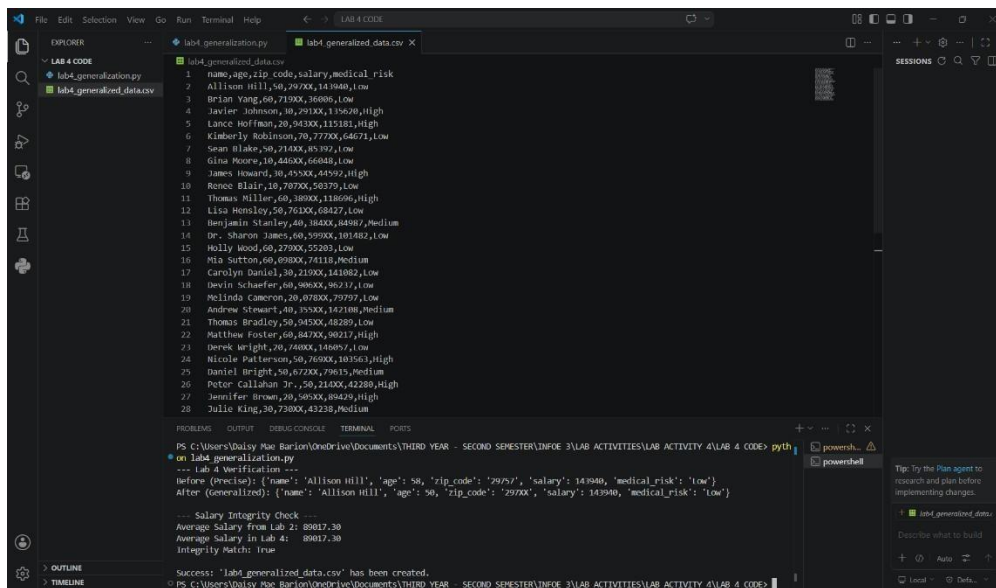
All ages in the CSV end in 0

All zip codes follow the format 123XX.



```
1 name,age,zip_code,salary,medical_risk
2 Allison Hill,50,29700,14990,Low
3 Brian Yang,60,71900,3600,Low
4 Javier Johnson,30,29100,13560,High
5 Lance Hoffman,20,94300,11518,High
6 Kimberly Robinson,70,77700,64671,Low
7 Sean Blake,50,21400,85392,Low
8 Gina Moore,10,44600,66048,Low
9 James Howard,30,45500,44592,High
10 Renee Blair,10,70700,50379,Low
11 Thomas Miller,60,38900,118696,High
12 Lisa Hensley,50,70300,68427,Low
13 Benjamin Stanley,40,18400,84987,Medium
14 Dr. Sharon James,60,59900,101482,Low
15 Holly Wood,60,27900,25283,Low
16 Mia Sutton,60,69800,74118,Medium
17 Carolyn Daniel,30,21900,141882,Low
18 Devin Schaefer,60,90000,96237,Low
19 Melinda Cameron,20,07800,79797,Low
20 Andrew Stewart,40,35500,142188,Medium
21 Thomas Bradley,50,94500,48289,Low
22 Matthew Foster,60,84700,90217,High
23 Derek Wright,20,74800,146057,Low
24 Nicole Patterson,50,76900,103563,High
25 Daniel Wright,50,67200,79615,Medium
26 Peter Callahan Jr.,50,21400,42280,High
27 Jennifer Brown,20,50500,89429,High
28 Julie King,30,73800,43238,Medium
29 Alexandra Lee,20,80000,45302,Low
30 Nicole Mack,70,45800,107736,Medium
31 Jeffrey Chavez,30,95100,131327,High
32
```

The Average Salary (from Lab 2) still matches, proving you generalized the people but didn't touch the sensitive financial data.



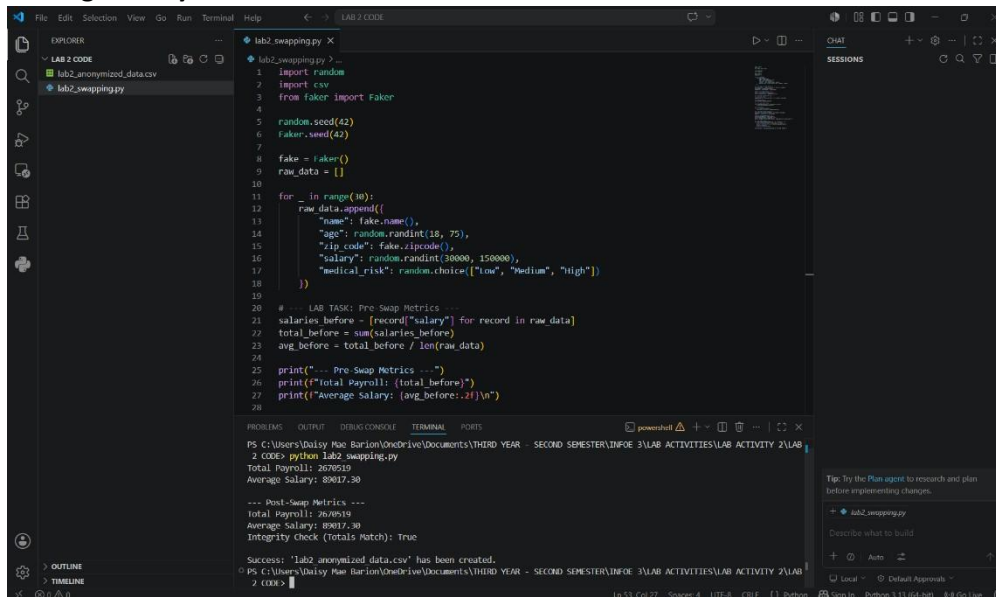
```
1 name,age,zip_code,salary,medical_risk
2 Allison Hill,50,29700,14990,Low
3 Brian Yang,60,71900,3600,Low
4 Javier Johnson,30,29100,13560,High
5 Lance Hoffman,20,94300,11518,High
6 Kimberly Robinson,70,77700,64671,Low
7 Sean Blake,50,21400,85392,Low
8 Gina Moore,10,44600,66048,Low
9 James Howard,30,45500,44592,High
10 Renee Blair,10,70700,50379,Low
11 Thomas Miller,60,38900,118696,High
12 Lisa Hensley,50,70300,68427,Low
13 Benjamin Stanley,40,18400,84987,Medium
14 Dr. Sharon James,60,59900,101482,Low
15 Holly Wood,60,27900,25283,Low
16 Mia Sutton,60,69800,74118,Medium
17 Carolyn Daniel,30,21900,141882,Low
18 Devin Schaefer,60,90000,96237,Low
19 Melinda Cameron,20,07800,79797,Low
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23 Derek Wright,20,74800,146057,Low
24 Nicole Patterson,50,76900,103563,High
25 Daniel Wright,50,67200,79615,Medium
26 Peter Callahan Jr.,50,21400,42280,High
27 Jennifer Brown,20,50500,89429,High
28 Julie King,30,73800,43238,Medium
29 Alexandra Lee,20,80000,45302,Low
30 Nicole Mack,70,45800,107736,Medium
31 Jeffrey Chavez,30,95100,131327,High
32
```

```
PS C:\Users\Daisy Mae Barion\OneDrive\Documents\THIRD YEAR - SECOND SEMESTER\INFOE 3\LAB ACTIVITIES\LAB ACTIVITY 4\LAB 4 CODE> python lab4_verification.py
-- Lab 4 Verification --
Before (Devised): {'name': 'Allison Hill', 'age': 50, 'zip_code': '29700', 'salary': 14990, 'medical_risk': 'Low'}
After (Generalized): {'name': 'Allison Hill', 'age': 50, 'zip_code': '29700', 'salary': 14990, 'medical_risk': 'Low'}

-- Salary Integrity Check --
Average Salary from Lab 2: 89017.30
Average Salary in Lab 4: 89017.30
Integrity Match: True

Success: 'lab4_generalized_data.csv' has been created.
PS C:\Users\Daisy Mae Barion\OneDrive\Documents\THIRD YEAR - SECOND SEMESTER\INFOE 3\LAB ACTIVITIES\LAB ACTIVITY 4\LAB 4 CODE>
```

Average Salary of Lab 2



```
lab2_swapping.py
1 import random
2 import csv
3 from faker import Faker
4
5 random.seed(42)
6 faker = Faker()
7
8 fake = faker()
9 raw_data = []
10
11 for _ in range(10):
12     raw_data.append({
13         "name": fake.name(),
14         "age": random.randint(18, 75),
15         "zip_code": fake.zipcode(),
16         "salary": random.randint(30000, 150000),
17         "medical_risk": random.choice(["low", "medium", "high"])
18     })
19
20 # --- LAB TASK: Pre-Swap Metrics ---
21 salaries_before = [record["salary"] for record in raw_data]
22 total_before = sum(salaries_before)
23 avg_before = total_before / len(raw_data)
24
25 print("--- Pre-Swap Metrics ---")
26 print(f"Total Payroll: {total_before}")
27 print(f"Average Salary: {avg_before:.2f}\n")
28
```

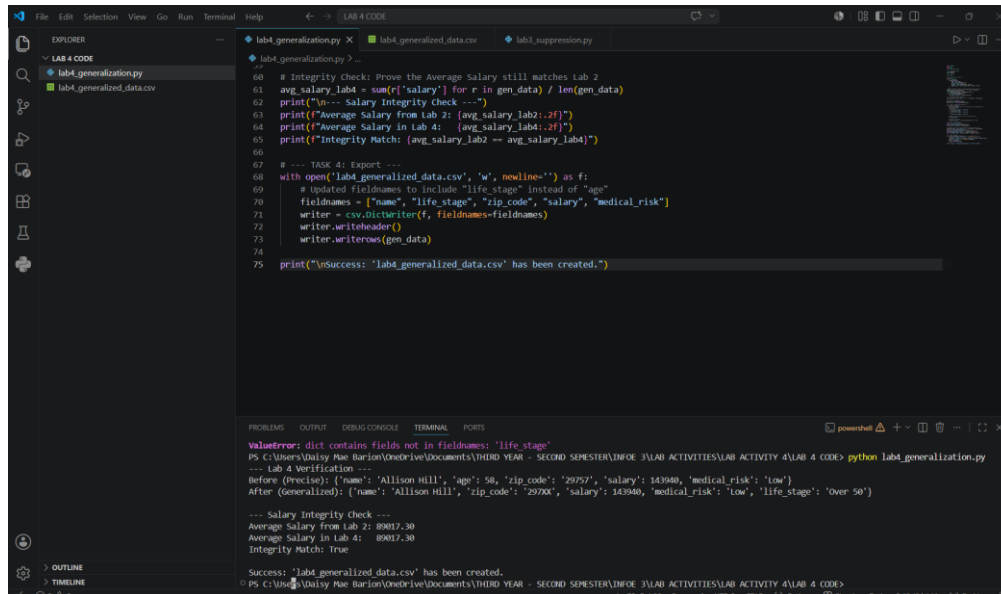
PS C:\Users\Daisy Mae Barion\OneDrive\Documents\THIRD YEAR - SECOND SEMESTER\INFO 3\LAB ACTIVITIES\LAB ACTIVITY 2\LAB 2 CODE> python lab2_swapping.py

Total Payroll: 2670519
Average Salary: 89017.30

--- Post-Swap Metrics ---
Total Payroll: 2670519
Average Salary: 89017.30
Integrity Check (Totals Match): True

Success: 'lab2_anonymized_data.csv' has been created.

Extra: Life Stages



```
lab4_generalization.py
60 # Integrity Check: Prove the Average Salary still matches Lab 2
61 avg_salary_lab4 = sum(r["salary"] for r in gen_data) / len(gen_data)
62 print("\n--- Salary Integrity Check ---")
63 print(f"Average salary from Lab 2: {avg_salary_lab2:.2f}")
64 print(f"Average salary in Lab 4: {avg_salary_lab4:.2f}")
65 print(f"Integrity Match: {avg_salary_lab2 == avg_salary_lab4}")
66
67 # --- TASK 4: Export ---
68 with open('lab4_generalized_data.csv', 'w', newline='') as f:
69     # updated fieldnames to include "life stage" instead of "age"
70     fieldnames = ["name", "life_stage", "zip_code", "salary", "medical_risk"]
71     writer = csv.DictWriter(f, fieldnames=fieldnames)
72     writer.writeheader()
73     writer.writerows(gen_data)
74
75 print("\nSuccess: 'lab4_generalized_data.csv' has been created.")
```

PS C:\Users\Daisy Mae Barion\OneDrive\Documents\THIRD YEAR - SECOND SEMESTER\INFO 3\LAB ACTIVITIES\LAB ACTIVITY 4\LAB 4 CODE> python lab4_generalization.py

ValueError: dict contains fields not in fieldnames: 'life stage'

--- Lab 4 Verification ---
Before (read): {'name': 'Allison Hill', 'age': 56, 'zip_code': '29752', 'salary': 143940, 'medical_risk': 'low'}
After (generalized): {'name': 'Allison Hill', 'zip_code': '29700', 'salary': 143940, 'medical_risk': 'low', 'life_stage': 'over 50'}

--- Salary Integrity Check ---
Average Salary from Lab 2: 89017.30
Average Salary in Lab 4: 89017.30
Integrity Match: True

Success: 'lab4_generalized_data.csv' has been created.

Short Description:

In Lab 4, we applied Generalization to quasi-identifiers to protect the dataset from linking attacks. Using floor division, we rounded ages to the nearest decade, and using string slicing, we masked the specific digits of zip codes. This transformation ensures that individual records are hidden within broader groups while the Average Salary remains perfectly accurate for statistical analysis.